

In [22]:

```
import pandas as pd
import numpy as np
from matplotlib import pyplot as plt
import seaborn as sns
```

In [96]:

```
df = pd.read_csv("primary/LUAD.clin.merged.picked.txt", sep="\t")
df.head()
```

Out[96]:

	Hybridization REF	tcga-05-4245	tcga-05-4382	tcga-05-4384	tcga-05-4396	tcga-05-4402	tcga-05-4405	tcga-05-4410	tcga-05-4418	tcga-05-4427	...	tcga-nj-a4yg
0	Composite Element REF	value	value	value	value	value	value	value	value	value	...	value
1	years_to_birth	81	68	66	76	57	74	62	69	65	...	65
2	vital_status	0	0	0	1	1	0	0	1	0	...	0
3	days_to_death	NaN	NaN	NaN	303	244	NaN	NaN	274	NaN	...	NaN
4	days_to_last_followup	730	607	426	NaN	NaN	610	0	NaN	791	...	2261

5 rows × 523 columns

In [97]:

```
df2 = pd.read_csv("primary/All_CDEs.txt", sep="\t")
df2.head()
```

Out[97]:

	bcr_patient_barcode	tcga-05-4245	tcga-05-4382	tcga-05-4384	tcga-05-4396	tcga-05-4402
0	bcr_patient_uuid	03d09c05-49ab-4ba6-a8d7-e7ccf71fafd2	3434b91a-c05f-460f-a078-7b1bb6e7085d	9a50e7e4-831d-489f-87d2-979e987561cc	9f81c602-8afa-4588-b0b6-6e5a1a128d5a	722d6818-f1cb-49f4-b68a-fd0a31304681
1	patient_id	4245	4382	4384	4396	4402
2	vital_status	alive	alive	alive	dead	dead
3	days_to_death	NaN	NaN	NaN	303	244
4	days_to_last_followup	730	607	426	NaN	NaN

5 rows × 523 columns

In [7]:

```
df2["bcr_patient_barcode"].unique()
```

Out[7]:

```
array(['bcr_patient_uuid', 'patient_id', 'vital_status', 'days_to_death',
      'days_to_last_followup', 'additional_studies',
      'age_at_initial_pathologic_diagnosis',
      'anatomic_neoplasm_subdivision',
      'anatomic_neoplasm_subdivision_other', 'b_symptoms', 'bcr',
      'bcr_canonical_reason-2', 'bcr_canonical_reason-3',
      'bcr_canonical_reason', 'bcr_followup_barcode',
      'bcr_followup_uuid', 'bcr_patient_canonical_status', 'clinical_
m',
      'clinical_n', 'clinical_stage', 'clinical_t', 'day_of_dcc_upload',
d',
      'day_of_form_completion', 'days_to_birth', 'days_to_index',
      'days_to_initial_pathologic_diagnosis', 'days_to_last_known_alive',
ve',
      'days_to_psa', 'diagnosis', 'dlco_predictive_percent', 'drugs',
      'eastern_cancer_oncology_group', 'egfr_mutation_identified',
      'egfr_mutation_performed', 'egfr_mutation_result',
      'eml4_alk_translocation_identified',
      'eml4_alk_translocation_method',
      'eml4_alk_translocation_performed',
      'eml4_alk_translocation_result', 'ethnicity',
      'extranodal_involvement', 'file_uuid', 'follow_ups',
      'followup_case_report_form_submission_reason',
      'followup_treatment_success', 'gender', 'gleason_score',
      'histological_type', 'history_of_neoadjuvant_treatment',
      'hiv_status', 'icd_10', 'icd_o_3_histology', 'icd_o_3_site',
      'igcccg_stage', 'informed_consent_verified',
      'karnofsky_performance_score', 'kras_gene_analysis_performed',
      'kras_mutation_found', 'kras_mutation_result',
      'location_in_lung_parenchyma', 'lost_follow_up', 'masaoka_stage',
e',
      'month_of_dcc_upload', 'month_of_form_completion',
      'number_pack_years_smoked', 'other_dx', 'pathologic_m',
      'pathologic_n', 'pathologic_stage', 'pathologic_t',
      'performance_status_scale_timing', 'person_neoplasm_cancer_status',
us',
      'post_bronchodilator_fev1_fvc_percent',
      'post_bronchodilator_fev1_percent',
      'pre_bronchodilator_fev1_fvc_percent',
      'pre_bronchodilator_fev1_percent', 'primary_pattern',
      'primary_therapy_outcome_success', 'progression_determined_by-
3',
      'psa_value', 'pulmonary_function_test_performed', 'race',
      'radiation_therapy', 'radiation_type_notes', 'radiations',
      'residual_tumor', 'secondary_pattern', 'serum_markers',
      'stopped_smoking_year', 'system_version',
      'targeted_molecular_therapy', 'tertiary_pattern',
      'tissue_prospective_collection_indicator',
      'tissue_retrospective_collection_indicator', 'tissue_source_site',
e',
      'tobacco_smoking_history', 'tumor_tissue_site', 'withdrawn',
      'year_of_dcc_upload', 'year_of_form_completion',
      'year_of_initial_pathologic_diagnosis',
      'year_of_tobacco_smoking_onset'], dtype=object)
```

In [89]:

```
df2t = df2.set_index("bcr_patient_barcode").transpose()
df2t.head()
```

Out[89]:

bcr_patient_barcode	bcr_patient_uuid	patient_id	vital_status	days_to_death	days_to_last_followu
tcga-05-4245	03d09c05-49ab-4ba6-a8d7-e7ccf71fafd2	4245	alive	NaN	73
tcga-05-4382	3434b91a-c05f-460f-a078-7b1bb6e7085d	4382	alive	NaN	60
tcga-05-4384	9a50e7e4-831d-489f-87d2-979e987561cc	4384	alive	NaN	42
tcga-05-4396	9f81c602-8afa-4588-b0b6-6e5a1a128d5a	4396	dead	303	Na
tcga-05-4402	722d6818-f1cb-49f4-b68a-fd0a31304681	4402	dead	244	Na

5 rows × 102 columns

In [90]:

```
df2t_stage = df2t[df2t["pathologic_stage"].isin(["stage i", "stage ia", "stage ib",
df2t_stage.head()
```

Out[90]:

bcr_patient_barcode	bcr_patient_uuid	patient_id	vital_status	days_to_death	days_to_last_followu
tcga-05-4245	03d09c05-49ab-4ba6-a8d7-e7ccf71fafd2	4245	alive	NaN	73
tcga-05-4382	3434b91a-c05f-460f-a078-7b1bb6e7085d	4382	alive	NaN	60
tcga-05-4384	9a50e7e4-831d-489f-87d2-979e987561cc	4384	alive	NaN	42
tcga-05-4396	9f81c602-8afa-4588-b0b6-6e5a1a128d5a	4396	dead	303	Na
tcga-05-4402	722d6818-f1cb-49f4-b68a-fd0a31304681	4402	dead	244	Na

5 rows × 102 columns

In [93]:

```
sorted(list(df2t_stage["pathologic_stage"].unique()))
```

Out[93]:

```
['stage i',
 'stage ia',
 'stage ib',
 'stage ii',
 'stage iia',
 'stage iib',
 'stage iiia',
 'stage iiib',
 'stage iv']
```

In [47]:

```
df2t_stage.groupby(by=["pathologic_stage", "gender"])["bcr_patient_uuid"].count().re
```

Out[47]:

	pathologic_stage	gender	bcr_patient_uuid
0	stage i	female	5
1	stage ia	female	83
2	stage ia	male	51
3	stage ib	female	75
4	stage ib	male	65
5	stage ii	female	1
6	stage iia	female	21
7	stage iia	male	29
8	stage iib	female	34
9	stage iib	male	39
10	stage iiia	female	40
11	stage iiia	male	34
12	stage iiib	female	6
13	stage iiib	male	5
14	stage iv	female	12
15	stage iv	male	14

In [59]:

```

_gender = df2t_stage.groupby(by=["pathologic_stage", "gender"])[ "bcr_patient_uuid" ].
_gender[ 'Percentage' ] = 100 * p_gender[ 'count' ] / p_gender[ 'count' ].sum()
rint(p_gender)
ns.set()
_gender_female = p_gender[p_gender[ "gender" ]=="female"]
_gender_male = p_gender[p_gender[ "gender" ]=="male"]

lt.figure(figsize=(12, 8))
lt.bar(p_gender_female["pathologic_stage"], p_gender_female["Percentage"], color='r')
lt.bar(p_gender_male["pathologic_stage"], p_gender_male["Percentage"], color='b')
lt.legend(("female", "male"))
lt.title("Gender percentage")
lt.xticks(rotation=45)

```

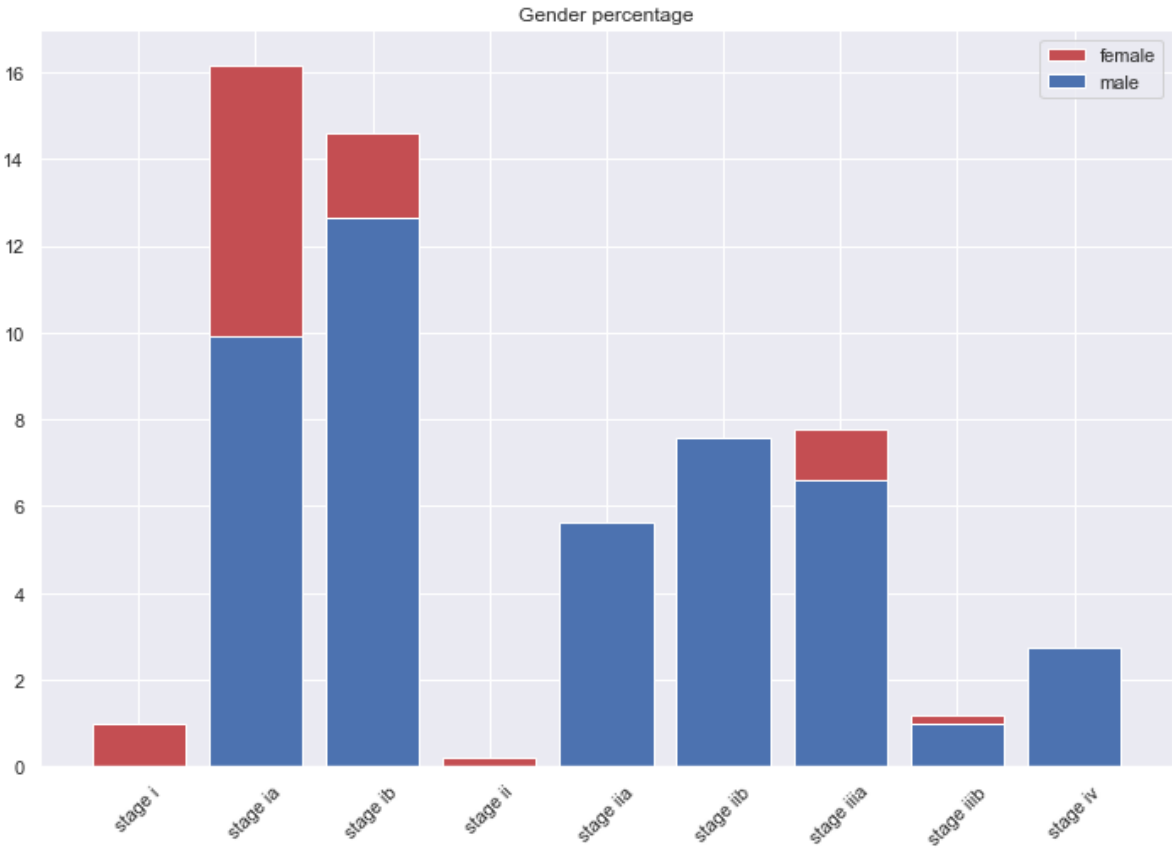
	pathologic_stage	gender	count	Percentage
0	stage i	female	5	0.972763
1	stage ia	female	83	16.147860
2	stage ia	male	51	9.922179
3	stage ib	female	75	14.591440
4	stage ib	male	65	12.645914
5	stage ii	female	1	0.194553
6	stage iia	female	21	4.085603
7	stage iia	male	29	5.642023
8	stage iib	female	34	6.614786
9	stage iib	male	39	7.587549
10	stage iiaa	female	40	7.782101
11	stage iiaa	male	34	6.614786
12	stage iiib	female	6	1.167315
13	stage iiib	male	5	0.972763
14	stage iv	female	12	2.334630
15	stage iv	male	14	2.723735

Out[59]:

```

([0, 1, 2, 3, 4, 5, 6, 7, 8], <a list of 9 Text xticklabel objects>)

```



In [61]:

```

p_race = df2t_stage.groupby(by=["pathologic_stage", "race"])[ "bcr_patient_uuid"].count()
p_race['Percentage'] = 100 * p_race['count'] / p_race['count'].sum()
print(p_race)
sns.set()
p_race_white = p_race[p_race["race"]=="white"]
p_race_asian = p_race[p_race["race"]=="asian"]
p_race_baa = p_race[p_race["race"]=="black or african american"]
p_race_aian = p_race[p_race["race"]=="american indian or alaska native"]

plt.figure(figsize=(12, 8))
plt.bar(p_race_white["pathologic_stage"], p_race_white["Percentage"], color='r')
plt.bar(p_race_asian["pathologic_stage"], p_race_asian["Percentage"], color='g')
plt.bar(p_race_baa["pathologic_stage"], p_race_baa["Percentage"], color='b')
plt.bar(p_race_aian["pathologic_stage"], p_race_aian["Percentage"], color="orange")
plt.legend(("white", "asian", "black or african american", "american indian or alaska native"))
plt.title("Race percentage")
plt.xticks(rotation=45)

```

	pathologic_stage	race	count	Percentage
0	stage i	black or african american	1	0.2237
14	stage i	white	4	0.8948
55	stage ia	asian	2	0.4474
27	stage ia	black or african american	18	4.0268
3	stage ia	white	110	24.6085
46	stage ia	white	110	24.6085
4	stage ib	asian	3	0.6711
01	stage ib	black or african american	8	1.7897
5	stage ib	white	102	22.8187
41	stage ii	black or african american	1	0.2237
6	stage iia	black or african american	9	2.0134
09	stage iia	white	39	8.7248
7	stage iia	white	39	8.7248
92	stage iib	american indian or alaska native	1	0.2237
8	stage iib	asian	1	0.2237
14	stage iib	black or african american	5	1.1185
9	stage iib	white	53	11.8568
23	stage iia	black or african american	7	1.5659
10	stage iia	white	55	12.3042
32	stage iia	white	55	12.3042
11	stage iib	white	7	1.5659
14	stage iib	white	7	1.5659
13	stage iib	white	7	1.5659
68	stage iib	white	7	1.5659
14	stage iib	white	7	1.5659
23	stage iia	black or african american	7	1.5659
15	stage iia	white	55	12.3042
96	stage iia	white	55	12.3042
16	stage iia	white	55	12.3042
51	stage iia	white	55	12.3042
17	stage iia	white	55	12.3042
96	stage iia	white	55	12.3042
18	stage iv	asian	2	0.4474
27	stage iv	asian	2	0.4474

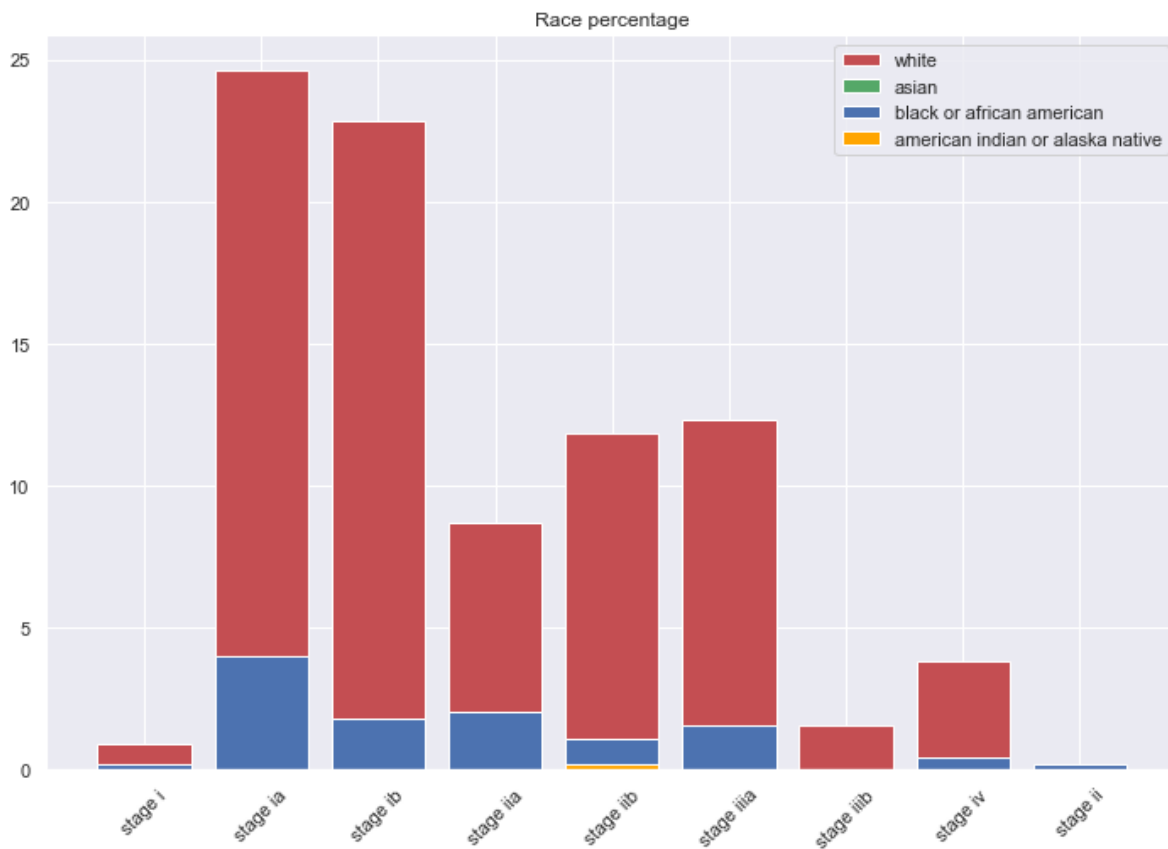
```

19          stage iv          black or african american          2          0.4474
27
20          stage iv                                white          17          3.8031
32

```

Out[61]:

([0, 1, 2, 3, 4, 5, 6, 7, 8], <a list of 9 Text xticklabel objects>)



In [83]:

```

p_both = df2t_stage.groupby(by=["pathologic_stage", "gender", "race"])[ "bcr_patient_
p_both['Percentage'] = 100 * p_both['count'] / p_both['count'].sum()
p_both[["gender", "race"]].drop_duplicates().values

```

Out[83]:

```

array([[ 'female', 'black or african american'],
       [ 'female', 'white'],
       [ 'female', 'asian'],
       [ 'male', 'black or african american'],
       [ 'male', 'white'],
       [ 'male', 'asian'],
       [ 'female', 'american indian or alaska native']], dtype=object)

```


In [86]:

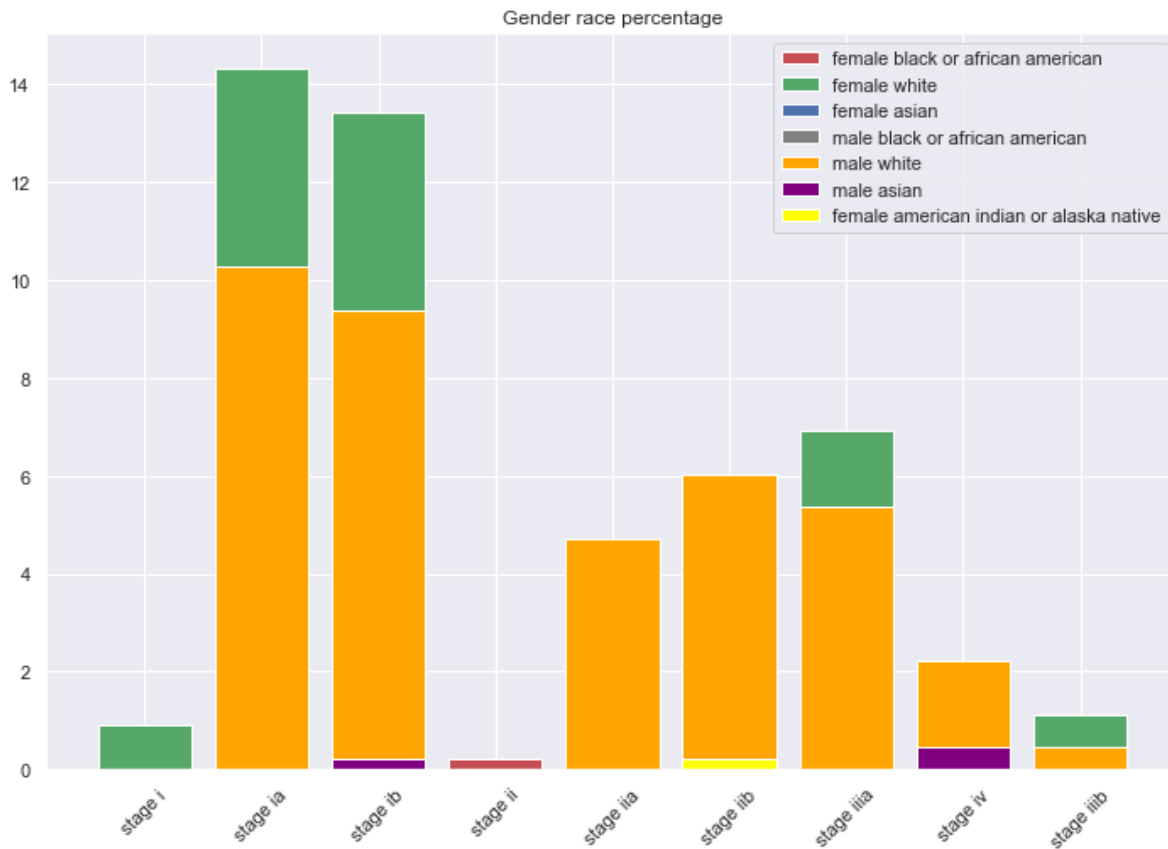
```
p_both = df2t_stage.groupby(by=["pathologic_stage", "gender", "race"])[ "bcr_patient_
p_both['Percentage'] = 100 * p_both['count'] / p_both['count'].sum()

plt.figure(figsize=(12, 8))
labels = []
colors = ["r", "g", "b", "grey", "orange", "purple", "yellow"]
for idx, [_gender, _race] in enumerate(p_both[["gender", "race"]].drop_duplicates()):
    _df = p_both[(p_both["gender"]==_gender) & (p_both["race"]==_race)]
    plt.bar(_df["pathologic_stage"], _df["Percentage"], color=colors[idx])
    labels.append(f"{_gender} {_race}")

plt.legend(labels)
plt.title("Gender race percentage")
plt.xticks(rotation=45)
```

Out[86]:

([0, 1, 2, 3, 4, 5, 6, 7, 8], <a list of 9 Text xticklabel objects>)



In [95]:

```
p_both.to_csv("p_both.csv")
```

In []: